# Fork-Based DeFi Execution Research: A Read-Only Methodology for Discovering, Validating, and Killing Execution Edges

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## Abstract

This paper presents a **fork-based, read-only methodology** for discovering, validating, and — most often — *killing* execution edges in decentralized finance (DeFi). Using deterministic Anvil forks over point-in-time (PIT) chain state, a full net-of-cost model, de-confounding, out-of-sample (OOS) testing, and tail stress, it drives every hypothesis to one of three verdicts: **FOUND** or **CLOSED**. Applied across oracle-lag liquidation, decentralized-exchange (DEX) and maximal-extractable-value (MEV) route economics, derivatives volatility-risk-premium (VRP), and cross-venue price-discovery flow, the methodology reproduces observed competitor economics yet shows that the **live-trading edges accessible to a single, public-data-only operator are market- and latency-bound and CLOSED**. The durable value is (a) a rigorous **negative-knowledge registry** and (b) **measurement products** (execution-quality, MEV-loss, oracle-freshness, liquidation-risk), with one **open-flow** candidate — a cross-venue flow signal that is not yet CLOSED (**centralized-exchange-to-DEX, "CEX→DEX", lead-lag**). This paper argues that in transparent, adversarial markets a rigorously-evidenced **CLOSED** verdict is the *expected and valuable* outcome of most edge research — not a failure but the result that prevents wasted capital — and that the durable, compounding asset such research yields is validated *measurement*, not a transient retained trading edge.

**Keywords:** DeFi, MEV, fork simulation, oracle lag, liquidation, execution quality, negative results, reproducibility.

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## 1. Introduction

Public blockchains make prices and pending flow broadly observable, so *prediction* is rarely the edge; **execution** is — who sees state first, who orders transactions, who pays for inclusion. This paper documents a disciplined program to test that premise empirically without risking capital: every candidate edge is measured on a **deterministic fork** of mainnet state and graded net of realistic cost. The contribution is not a single profitable strategy but a **reproducible method** and a body of **honest verdicts**, including the finding that the most promising live edge (oracle-lag liquidation) is real in research but **market-bound in live execution** for a single, public-data-only operator.

### 1.1 Verdict taxonomy and key terms

Every candidate surface is driven to exactly one verdict; these labels are used consistently throughout the paper and are defined here once.

- **CONFIRMED (research).** The mechanism and its magnitude are reproduced on fork/read-only evidence. It says the effect is *real*, not that it is *tradeable* by this operator (which is a separate, cost- and market-bound question).
- **FOUND.** An edge that survives the full evidence ladder of Section 3 — de-confounding, out-of-sample (OOS) testing, tail stress, **and** the net-of-cost model — and is therefore judged a real, operator-actionable trading edge. No surface in this paper has reached FOUND.
- **FOUND_CANDIDATE.** A signal that is reproducibly present (e.g. survives de-confounding and repeats across sessions) but has **not yet** cleared the net-of-cost feasibility test for a specific monetization. It is a live hypothesis, not a confirmed edge.
- **PAPER_CANDIDATE.** Positive in live-data paper/shadow testing, pending a full production-path verdict; carries no live capital.
- **CLOSED.** Rejected as not operator-actionable, with a measured reason and a precise reopen condition. Qualified variants record *why*:
  - **CLOSED_BELOW_COST** — gross-positive but net-negative once swap fees, gas, price impact, and slippage are subtracted.
  - **CLOSED_NO_EDGE** — no exploitable gross dislocation (or no OOS-stable signal) in the first place.
  - **CLOSED_NO_SECONDARY_MARKET** — structurally unexecutable because the token trades on a single venue, so no two-venue route exists.
  - **CLOSED (market)** — the mechanism is real and the software is fast enough, but too few qualifying opportunities exist for a public-data-only operator to win inclusion (a market/visibility constraint, not a latency or code constraint).
- **Open flow (open-flow candidate).** A cross-venue order-/price-flow surface whose signal is reproducibly present (FOUND_CANDIDATE) but whose net-of-cost monetization has **not yet been established** — *not yet CLOSED* and not yet FOUND; its resolution is left to future work. The program has exactly one (Section 8).

## 2. Related work

This methodology sits at the intersection of three literatures. **Market microstructure** (Kyle 1985; Glosten–Milgrom 1985; Hasbrouck 1991; O'Hara 1995) explains how order flow, information asymmetry, and immediacy divide surplus between liquidity takers and providers. **Blockchain value extraction** — Daian et al.'s *Flash Boys 2.0* (2020) on MEV and priority-gas auctions, the proposer–builder separation (PBS) and relay literature, and Milionis et al. (2022) on loss-versus-rebalancing (LVR) — characterises how ordering rights are auctioned and how automated-market-maker (AMM) liquidity providers are adversely selected. **Optimal execution and transaction-cost analysis** (Almgren–Chriss 2000) formalises the impact–delay trade-off behind the cost model.

**Relation to recent empirical and tooling work.** Historical profitability measurement, automated DeFi opportunity discovery, fork-based counterfactual experimentation, and the decisive role of execution costs and operational constraints are all established in prior work; this paper therefore claims novelty in none of them. Empirical MEV quantification is developed by Qin, Zhou and Gervais (2022) and extended by Chi, He, Hu and Wang (2024), Piet, Fairoze and Weaver (2022), and Weintraub, Ferreira Torres, Nita-Rotaru and State (2022); automated profit-generating-transaction discovery by Zhou, Qin, Cully, Livshits and Gervais (2021) (DEFIPOSER); historical arbitrage analysis and candidate execution by the goldphish analyzer; fork-based counterfactual testing is demonstrated in practitioner work (Chen, 2026); and Schwertfeger and Vogt (2026), using a live high-frequency system, show that operational costs, liquidity constraints, market impact, competi-

tion, and blockchain execution risk sharply limit realized cross-venue arbitrage relative to frictionless backtests. Against this base, the contribution here is the **systematic use of these established techniques to maintain an auditable, candidate-level negative-knowledge registry** — driving every candidate surface to a FOUND / CLOSED verdict with an explicit reopen condition — together with **operator-frontier conditioning** (public-data-only, pruned node, single machine) that separates "no edge exists" from "no edge *for this operator*", and an honest single-operator live result diagnosing a confirmed edge as market- rather than latency-bound.

## 3. Methodology — fork-based measurement (the evidence ladder)

1. **Observe** — read-only public data; pin block/tx/log-index; reconcile clocks (event vs block vs ingest); record provenance (value, schema, source).
2. **Simulate** — re-execute on an Anvil fork of PIT state; determinism guarantees identical results and auditability. Fast analytic AMM/route models are validated against fork replay.
3. **Cost model** — subtract swap fees, gas × congestion-aware price, price impact ($\sqrt{}$size), slippage, funding, and failed-attempt gas. Gross-positive but net-negative ⇒ CLOSED_BELOW_COST.
4. **De-confound** — orthogonalize against BTC beta and common drivers; permutation/placebo nulls.
5. **OOS + multiple testing** — forward-chaining cross-validation (CV); deflated Sharpe; block permutations for autocorrelated series.
6. **Tail stress** — inject LUNA/FTX/COVID-scale gaps; confirm no single tail breaches the risk budget.
7. **Paper → shadow** — live-data paper, then full production path with the sender disabled and fail-closed.
8. **Verdict + record** — FOUND / CLOSED, recorded in an auditable registry.

**Resource frontier.** Verdicts are conditioned on the operator's envelope: no archive node, no colocation/sequencer adjacency, no trading capital or venue accounts, public-data-only. This frontier (detailed in Appendix A) separates "no edge exists" from "no edge *for this operator*."

## 4. Summary of measured results

The methodology of Section 3 (the eight-step evidence ladder) and the data-processing procedure of Appendix A were applied to each candidate surface. Table 1 consolidates the resulting headline measurements and verdicts; the per-surface detail follows in Sections 5–8. All reported results are fork-measured or read-only, graded net of the full cost model, and carry the resource-frontier and public-data caveats of Section 3 and Appendix A.

**Table 1. Measured results and verdicts by surface (read-only; fork-measured or public-data).**

| Surface | Measured quantity | Result | Verdict |
|---|---|---|---|
| Oracle-lag liquidation — research | positive-window rate (share of monitored oracle-freshness windows with true HF $< 1$ while the protocol still reads $\geq 1$); one on-chain competitor-liquidation receipt + a net-of-cost margin model | **28.9% of 720** windows positive (≈26–31% across runs); representative competitor liquidation **886,390 gas** (wstETH/USDe, block 25279838); **modeled net margin ≈5.29–5.84%** (600 bps bonus net of costs) | CONFIRMED (research) |
| Oracle-lag liquidation — live | 2,319 backrun submissions over ~4 weeks | 0 wins, 0 reverts, 0% outbid | CLOSED (market, not latency) |
| DEX/DEX bluechip backrun (WETH/USDC and other deep pairs) | gross displacement vs full cost stack | 11–17 basis points (bps) vs ~40 bps cost | CLOSED_BELOW_COST |
| Stablecoin routes | near-par pair arb (USDC/USDT, USDC/DAI): gross vs cost | 0.35–0.80 bps vs ~9 bps cost | CLOSED_BELOW_COST |
| Long-tail newly-launched V4 tokens (meme pairs) | on-chain V2/V3/V4 pool lookup across all fee tiers; rescan of ~1,000 recent blocks (61 non-bluechip tokens, 25 most-active analyzed) | 18/25 are V4-only (no secondary market); 7/25 have a V2/V3 market | CLOSED_NO_SECONDARY_MARKET (V4-only majority); the 7 with a secondary market → DEX/DEX CLOSED_BELOW_COST |
| Uniswap-V4 / universal-router backrun | 82 deterministic forks | 0 net-positive | CLOSED_NO_EDGE |
| Derivatives VRP / defined-risk iron-fly | forward implied-minus-realized volatility (IV−RV); testnet iron fly held to settlement | +3.3 percentage points | PAPER_CANDIDATE |

| Surface | Measured quantity | Result | Verdict |
|---|---|---|---|
| Carry, funding-spread, GMX open-interest (OI) imbalance, perp basis, options IV-lag | read-only CEX / Deribit data | carry: 0/400 high-funding periods; Hyperliquid/dYdX funding converged ~0 bps; GMX OI net −161/−191 bps annualized percentage rate (APR) after churn (repeated side-flip entry/exit cost); perp basis −1.7/−2.7 bps ≪ cost; IV-lag corr <0.30 at 1–60 s | CLOSED |
| CEX→DEX lead-lag | Binance-spot vs Uniswap-V3 timing over 51 volatile sessions | spot leads in 38/51 sessions, ≈6 s median lead; basis 1–4 bps ≪ ≈15–16 bps round-trip | FOUND_ CANDIDATE (direct arb CLOSED; directional / informed-LP net-of-cost feasibility **open — future work**) |

### 5. Oracle-lag liquidation (research CONFIRMED; live CLOSED — market)

**Mechanism.** Chainlink push-oracles lag DEX price discovery; during the lag a borrower's true health factor (HF) can be $< 1$ while the lending protocol still reads $\geq 1$. An oracle update then compels liquidation — a forced flow.

**Findings.** The lag is repeatedly confirmed — a positive window (true HF $< 1$ while the feed still reads $\geq 1$) in **28.9% of 720 monitored oracle-freshness windows** (≈26–31% across runs). The liquidation economics are anchored to on-chain evidence plus a net-of-cost model rather than a multi-event replay: a representative competitor liquidation of a wstETH/USDe Aave V3 position (**block 25279838**) used **886,390 gas** (on-chain receipt, status success), and the net margin is **modeled at ≈5.29–5.84%**, driven by the **6% (600 bps) Aave liquidation bonus** net of gas, the flash-loan fee, and the ≈21–57 bps oracle-to-DEX price gap (these describe third-party liquidation economics, not the operator's own trades). A full execution stack (extensively unit-tested) submits directly to permissionless builder endpoints over a fast software hot-path — sub-millisecond detection-to-submission, reaching the fastest builders within a few milliseconds, i.e. well inside the ≈12 s block — so inclusion speed is not the binding constraint.

**Live result.** Over **≈4 weeks and 2,319 backrun submissions: 0 wins, 0 reverts, 0 outbids.** Non-inclusions were either not-liquidatable or below a minimum-debt (dust) floor; an outbid rate of 0% over 2,319 attempts implies the constraint is **too few deep, non-dust HF<1 crossings in the monitored set**, not latency. **Verdict: live-trading edge CLOSED (market, not latency).** A representative near-miss was lost next-block to a competitor paying via private order flow off a prior-block crossing — a visibility/placement loss, not a bid/latency loss.

## 6. DEX/MEV route economics (CLOSED registry)

Fork-measured across surfaces: single-pool backrun (CLOSED_BELOW_COST), DEX/DEX bluechip (11–17 bps vs ~40 bps cost), **stablecoin routes** (below), **long-tail newly-launched V4 tokens** (below), and Uniswap-V4 / universal-router (UR) backrun (82 forks, 0 net-positive).

**These DEX/MEV surfaces were all fork- and shadow-measured, never traded with live capital.** The only surface taken to live submission in the broader project is oracle-lag liquidation (Section 5, 2,319 live backrun submissions on bluechip Aave collateral/debt assets); it is reported separately and is the single live-traded result. The coins exercised on the DEX/DEX surfaces fall into three groups: (i) **bluechips** — WETH, WBTC, and the major USD stablecoins in deep pools; (ii) **near-par stablecoin pairs** — USDC/USDT and USDC/DAI; and (iii) **non-bluechip long-tail tokens** — newly-launched Uniswap-V4 meme tokens (originally **BINS** and **MEETING**, extended by a rescan covering 61 such tokens). All three groups are CLOSED, but for distinct, measured reasons (below).

**Stablecoin routes** are arbitrages between near-par stablecoin pairs (e.g. USDC/USDT, USDC/DAI) across pools. Because both legs trade within a tight band around $1, the exploitable mispricing is tiny — **0.35–0.80 bps gross**, well below the ~9 bps round-trip cost (swap fees + gas) — so the surface is CLOSED_BELOW_COST.

**Long-tail newly-launched V4 tokens.** The first candidates were two V4-only meme tokens — **BINS** and **MEETING** — each with exactly one venue (recorded in the closed-track registry, Section 9). A broader read-only rescan (~1,000 recent mainnet blocks via a public archive endpoint) found **61 distinct non-bluechip tokens** appearing in newly-initialized V4 pools; among the **25 most-active**, **18 remained V4-only** (no V2 or V3 pool at any fee tier against WETH or USDC — e.g. ZC, PERCY, VERIDIA, APESHIT, SPIDERMAN) while **7 had a V2/V3 secondary market** (e.g. SHIB, WDOGE, OMNI, AXL). The V4-only majority is **CLOSED_NO_SECONDARY_MARKET** — structural, since no two-venue route exists. The minority with a secondary market is **not** structurally closed but collapses into the ordinary DEX/DEX result: the cross-venue spread on these thin, newly-launched pairs does not survive the ~40 bps-class cost stack, leaving a latency / order-flow race (CLOSED_BELOW_COST), and these were not individually fork-measured in this pass. Either way no operator-actionable edge appears; the V4-only surface reopens only if durable secondary liquidity for the same token appears and the cross-venue spread exceeds combined fees and gas. (Token identities were checked on-chain via the ERC-20 `symbol()`/`name()` methods: the established secondary-market examples — SHIB, OMNI, AXL — resolve to their canonical mainnet contracts, while the long-tail V4-only names are newly-deployed tokens with self-reported, uncurated tickers.)

Uniformly, the **gross displacement** — the raw price dislocation a swap leaves in a pool, measured *before* costs (i.e. the pre-cost size of the arbitrage) — is small relative to the DeFi cost stack. After subtracting swap fees, gas, and price impact, the residual is a **latency / order-flow race** that a single public-data-only operator cannot win. Sandwiching is analytics-only by ethical charter.

## 7. Derivatives (VRP paper-positive; carry/funding CLOSED)

Read-only Deribit / centralized-exchange (CEX) data. VRP / defined-risk iron-fly is PAPER_CANDIDATE (forward IV−RV +3.3 percentage points; testnet iron fly held to settlement). Carry, funding-spread (Hyperliquid / dYdX, GMX OI imbalance), perpetual funding-basis reversion, and options IV-lag are CLOSED, each with a measured reason: **carry** showed no signal in **0 of 400**

high-funding periods; the **Hyperliquid/dYdX funding spread** converged to ~**0 bps**; **GMX OI-imbalance** capture was not robust (**net −161/−191 bps APR** after churn — the repeated entry/exit cost incurred each time the imbalance side flips and the position must be reversed); **perpetual funding-basis reversion** was below cost (ETH/BTC basis **−1.7/−2.7 bps**, far under round-trip cost); and **options IV-lag** showed no persistent signal (cross-correlation **< 0.30 at 1–60 s** lags).

## 8. Cross-venue flow — the one open candidate

This is the program's single **open-flow** candidate (Section 1.1): a cross-venue surface whose signal is reproducibly present but whose net-of-cost monetization has not been established in this study.

**CEX→DEX lead-lag.** Across **51 volatile sessions, Binance spot led Uniswap V3 in 38 (≈75%)**, with a **median lead of ≈6 s** — a reproducible **FOUND_CANDIDATE**. Direct cross-venue arbitrage on this signal is nonetheless **CLOSED**: the measured cross-venue basis is only **1–4 bps**, far below the **≈15–16 bps round-trip cost** (two swaps at ≈7.6 bps each ≈ 5 bps pool fee + 2 bps slippage + 0.6 bps gas per leg). The one surface that is *not* yet closed is the **net-of-cost feasibility of a directional or informed-liquidity-provision (informed-LP) use** of the lead, rather than the direct race: a *directional* use would take a short-horizon DEX position in anticipation of the ≈6 s-lagged move and close once it realizes, graded net of swap fees, gas, slippage, and price impact; an *informed-LP* use would provide liquidity while using the lead to re-quote or withdraw just ahead of the predicted move, earning fees while avoiding the adverse-selection (LVR) loss that uninformed liquidity providers bear. Establishing that net-of-cost verdict requires a longer high-resolution sample of volatile sessions than this study captured.

## 9. Negative knowledge (the closed-track registry)

The program's largest output is a **canonical CLOSED registry**: every disproven surface, its measured reason, and a precise reopen condition. This prevents rework and, crucially, calibrates confidence in the few survivors — many independent CLOSED verdicts around one mechanism make the surviving signal more credible, not less.

## 10. From trading to measurement (the product pivot)

Because the operator-accessible trading edges are CLOSED, the durable value is **measurement**: transaction-audit, execution-quality, MEV-loss, oracle-freshness, and liquidation-risk reporting. As actually run, these were produced on **pruned full nodes (Erigon primary, Geth secondary) with deterministic Anvil forks over recent point-in-time state and public data only** — so MEV-loss is reported as a **conservative public-data lower bound** rather than an exact figure (exact reconstruction was not performed, as it needs an archive node the study did not have). These reuse the same fork/PIT machinery and require no capital; upgrading the lower bounds to *exact* figures would additionally require an archive node.

## 11. Limitations and ethics

**Limitations.** Public-data ceiling (no private orderflow); **no archive node** (blocks exact reconstruction and long-history backtests, so historical coverage is biased toward recent, pruned-node-reconstructable state); single machine, no colocation/sequencer adjacency (loses latency races, including layer-2 (L2) and Timeboost auctions); no trading capital or venue accounts (no live carry,

options, or market-making); small sample sizes and short windows widen confidence intervals; and geographic/ISP restrictions limit access to some venues. These bound the *actionability* of the results, not the validity of the measurements.

**Ethics.** The program is measurement-oriented and read-only: it quantifies extraction and its costs to inform market design and user protection, and it does not perform front-running, sandwiching, or censorship. No live capital is deployed for the research reported here; sandwiching is treated as an analytics-only subject by charter. The one live on-chain capability in the broader project is a solvency-improving liquidation service run under explicit human authorization, which is out of scope for this paper's claims.

## 12. Conclusion

A fork-based, read-only methodology lets a single operator honestly measure DeFi execution edges without risking capital. Applied here it yields mostly **well-evidenced CLOSED verdicts**, one confirmed-but-market-bound edge (oracle-lag liquidation), one open flow candidate (CEX→DEX), and a clear pivot toward **measurement products**. In transparent, adversarial markets, disciplined negative results and validated measurement are the reproducible asset.

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## Appendix A — Resource frontier and reproducibility

**Resource frontier.** Every verdict is conditioned on the operator's envelope, which separates "no edge exists" from "no edge *for this operator*": public data only (no private order flow); a **pruned full node** (Erigon primary, Geth secondary), *not* an archive node; a single machine with no colocation or sequencer adjacency; and no trading capital or venue accounts.

**Reproducibility.** The reported results are the entries of Table 1 and the per-surface numbers in Sections 5–8. Every such number traces to pinned finalized blocks (block, transaction, and log index), the exact code commit, parameters, and random seeds; monetary values use integer base-unit arithmetic with explicit token decimals, and prices distinguish spot, mark, index, and oracle pipelines. Fork replays are deterministic and therefore auditable. The research code and the pinned datasets already exist and underlie every reported number, and a **consolidated reproducibility manifest** — data window, commit hash, parameters, and random seeds, together with a per-surface index of the research code and datasets — has been assembled and will be **released with the submission**. Repository-specific paths and commit identifiers are recorded in that manifest, not in this paper. None of these artifacts are embedded in this draft, and any figures prepared for the **camera-ready version** — the final, peer-review-accepted manuscript formatted to the publication venue's template for printing/publication — are generated from the same pinned data. Author-identifying operational details are omitted for double-blind review.

## Author contributions

**Prakash Vishnoi** (sole author) conceived and designed the study, implemented and reviewed the research software and data-processing pipeline, performed and verified the measurements and analysis, and wrote and revised the manuscript, and takes full responsibility for its contents. Generative-AI tools were used as research and writing aids (see *Disclosure of generative-AI use*) and are **not** an author.

## Disclosure of generative-AI use

Generative-AI tools (large-language-model-based coding assistants) were used during this project for research- and writing-support tasks, specifically: implementation and modification of the measurement, simulation, and data-processing code; preparation of the tables; and drafting and language editing of this manuscript. AI tools were not used as a source of factual, empirical, or bibliographic claims. All AI-assisted material — code, computations, tables, and text — was reviewed and verified by the author(s), who independently checked the methodology, data-processing procedures, and reported results. No AI system is an author; the author(s) take full responsibility for the accuracy, integrity, methodology, results, and conclusions of this work.

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